Chemical Kinetics of Solids
by Hermann Schmalzried
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Epilogue

Maybe it is Aristotle (384–322 BC, "Corpora non agunt nisi fluida") who is responsible for the fact that chemists neglected the quantitative approach to solid state kinetics for such a long time. Certainly, however, it is the complexity of the kinetic problems which impeded, and still impedes, pertinent studies. Not only must the formal treatment (presented in textbooks on homogeneous/heterogeneous kinetics and transport) be mastered, but also the influence of elasticity, plasticity, crystallographic constraints, and the response of solids to electromagnetic fields during the course of solid state reactions.

The foregoing chapters mark a long and not yet finished journey through the special field of the chemical kinetics of solids. It differs from the more common textbooks on kinetics not only because of the immense variety of crystalline phases, but even more in view of the ambiguity in the definition of the correct number of independent thermodynamic state variables. This is the source of many difficulties and particularly with solids containing one or more immobile components or multiphase systems composed of coherent or semicoherent crystals. In coping with this inherent complexity in the foregoing chapters, we chose to restrict ourselves mainly to the fundamental aspects rather than to present many uncorrelated details.

In order to solve a kinetic problem (*i.e.*, the evolution in time of a non-equilibrium ensemble of mobile atomic particles), we have first to construct theoretically, or to determine experimentally, the Gibbs energy function of the system which we wish to investigate. From this potential, we can then derive the appropriate driving forces. Whereas in homogeneous systems we need to know affinities, in inhomogeneous systems we have to deduce the potential gradients, at least in the framework of the thermodynamics of (linear) irreversible processes. Secondly, it is necessary to understand the kinetic coefficients in the rate equations of solid state processes. These coefficients are closely connected to the dynamics of atomic motions, and to the type and number of crystal defects. As long as we use the concepts of thermodynamics of irreversible processes, we assume tacitly that local equilibrium prevails and, in particular, that point defect equilibrium is established during reaction. In such cases, each volume element in the reaction zone possesses a local Gibbs energy function which is fully determined by the set of independent thermodynamic state variables.

In many non-equilibrium situations, this local equilibrium assumption holds for the crystal bulk. However, its verification at the phase boundaries and interfaces (internal and external surfaces) is often difficult. This urges us to pay particular attention to the appropriate kinetic modeling of interfaces, an endeavour which is still in its infancy.

Thirdly, we note that a kinetic problem is never properly defined unless the boundary conditions are known and specified throughout. This becomes a critical point when boundaries move during the solid state reaction and their morphology is un-

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stable. Boundary morphologies can be influenced easily by the elastic properties and plastic behavior of contiguous phases.

This monograph is an attempt to quantify the complex kinetics of solids. It can be done on different levels of sophistication. Statistical theory requires atomic scale models which could be based on electron theory. One may also work on a less fundamental level by introducing phenomenological parameters. Yet even this latter approach is not always feasible for solids with more or less ordered systems having very large numbers of interacting atomic particles, despite the availability of modern computers. In practice, the linear thermodynamics of irreversible processes is a useful frame since transport is responsible for the larger part of solid state kinetics. We must remain aware, however, that the kinetic coefficients are *ad hoc* parameters, unless they can be derived from atomic theory [A. R. Allnatt and A. B. Lidiard: *Atomic Transport in Solids*]. Thus, a book on solid state kinetics is a compromise, by necessity, and often gives the definition of a problem rather than the solution. However, if the definition is correct and unique, one day we will have the unambiguous answer to the problem.

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